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**ATTA-UR-RAHMAN SCHOOL OF APPLIED BIOSCIENCES (ASAB)**
**& SCHOOL OF CHEMICAL AND MATERIALS ENGINEERING (SCME)**

# TECHNICAL BRIEFING REPORT
# DNA BASE TAG AND TRACE SYSTEM

**Indigenous Bio-Molecular Tagging for Strategic Industries**

**Date:** August 2026 | **Security Classification:** Restricted / Proprietary

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|---|---|
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Islamabad, Pakistan — 2026

## Project Approvals

This written technical brief documents the system architecture, operational workflow, mathematical calculations, and empirical qPCR verification of the **DNA Base Tag and Trace System**.

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# 1. Project Background & Objectives

## 1.1 Project Background

Counterfeiting, product adulteration, and illicit supply chain diversion represent severe challenges to national security, public health, and tax revenue. High-value sectors — including pharmaceuticals, defense materials, petroleum, agrochemicals, and excise goods — require an absolute authentication mechanism that cannot be cloned or reverse-engineered.

Conventional security features such as barcodes, optical holograms, and RFID tags are easily forged or detached from physical goods. To solve this problem, NUST developed the **DNA Base Tag and Trace System**. By using synthetic DNA molecules as invisible, unforgeable markers, any physical product can be uniquely identified and verified with forensic certainty.

## 1.2 Conventional Markers vs. DNA Tagging

Table 1.1 compares traditional anti-counterfeiting markers with the synthetic DNA tagging system.

**Table 1.1.** Comparison: Conventional Physical Security vs. Synthetic DNA Tagging

| Feature | Conventional Markers | DNA Tagging System |
|---|---|---|
| **Visibility** | Overt / Visible (Subject to duplication). | **Covert**: Invisible micro-traces (< 1 ng/L). |
| **Unforgeability** | Low (Subject to forgery). | **Absolute**: > $10^{120}$ theoretical sequence space. |
| **Durability** | Susceptible to heat and wear. | **High**: Encapsulated in silica/ceramic (> 800°C). |
| **Verification** | Visual / Optical scan. | **Forensic qPCR**: Quantitative optical detection. |

## 1.3 Project Objectives

The primary objectives of this project are:

1. **In-House Software Engine**: Build software for automated DNA sequence design, primer selection, and multiplex probe assignment.
2. **Physical Synthesis & Scaling**: Synthesize custom DNA constructs (such as the

200 bp prototype validated in this project), perform bulk PCR amplification, and formulate carrier buffers.

3. **Product Integration**: Deploy taggants onto consumer packaging, liquids, and security stamps using automated spray machinery.

4. **Forensic qPCR Verification**: Establish fast field recovery protocols and multi-channel qPCR assays to verify taggants with 100% certainty.

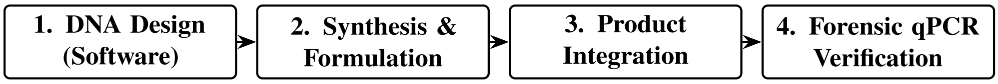


**Figure 1.1.** Overview of the DNA Tag and Trace workflow.

# 2. System Workflow

## 2.1 Workflow Overview

The DNA Tag and Trace workflow consists of five clear operational stages (Figure 2.1).

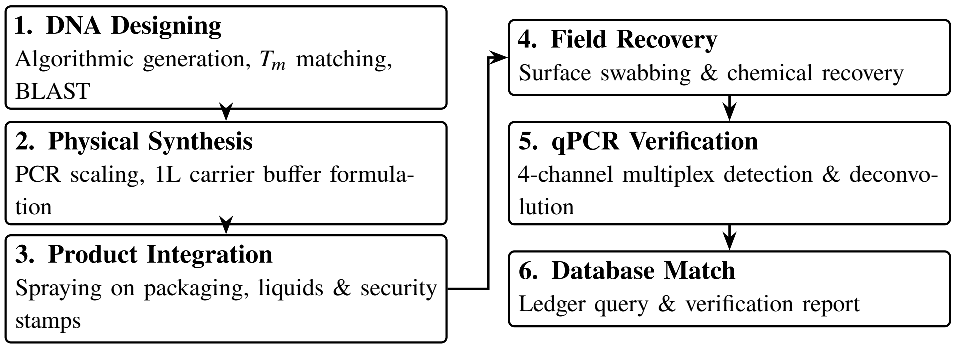


**Figure 2.1.** The five stages of the DNA tagging and verification lifecycle.

## 2.2 DNA Designing

DNA constructs are designed through the custom software engine:

- **Custom Sizing**: The software supports arbitrary, user-defined sequence lengths according to industrial application requirements (for this proof-of-concept validation, a 200 bp prototype construct was engineered and synthesized).
- **Structure**: Flanked by 20 bp forward and reverse primers, containing four internal 24 bp fluorescent probe targets (FAM, HEX, Texas Red, Cy5) separated by spacers (Figure 2.2).
- **Thermodynamic Boundaries**: Melting temperatures ($T_m$) are calculated using SantaLucia parameters:

$$T_m = \frac{\Delta H^\circ}{\Delta S^\circ + R \ln(C_T/4)} + 16.6 \log_{10}[\mathrm{Na}^+] \tag{2.1}$$

  Primers are matched at $T_m = 58.5 \pm 1.0$°C, and probes are set at $T_m \geq 68.0$°C (+10.0°C above primers).

- **Safety Screening**: Local BLAST checks ensure zero homology (< 25%) with any known environmental or human pathogens.

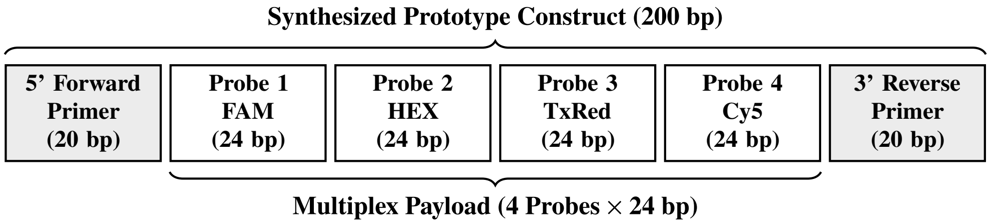


**Figure 2.2.** Molecular layout of the synthesized 200 bp prototype DNA taggant construct.

## 2.3 Physical Synthesis & Formulation

1. **Bacterial Host Propagation**: The cloned construct was supplied in transformed *Escherichia coli* (Stab Top10) on ampicillin-selective LB agar and cultured in LB broth at 37°C for 16 hours.

2. **Rapid Heat-Lysis Screening**: Individual bacterial colonies were suspended in 100 $\mu$L TE buffer, heat-treated at 95°C for 10 minutes in a thermal cycler, and briefly spun. The supernatant was used directly as PCR template, eliminating the need for commercial DNA extraction kits.

3. **Standard PCR Master Mix (**25 $\mu$**L Volume)**: Reagents were combined in 0.2 mL tubes comprising 2.5 $\mu$L 10× PCR buffer, 1.5 $\mu$L $MgCl_2$, 1.0 $\mu$L dNTPs, 1.0 $\mu$L Taq DNA polymerase, 1.0 $\mu$L each of forward and reverse primers, 1.0–2.0 $\mu$L template DNA, and 15.0–16.0 $\mu$L distilled water.

4. **Annealing Optimization & PCR Scaling**: Gradient PCR performed across 54°C–59°C identified 57.0°C as the optimal annealing temperature. 35 cycles of amplification produced pure 200 bp amplicon stocks.

5. **1.0-Liter Master Spray Formulation**: 1.0 Liter of stabilized TE buffer (10 mM Tris-HCl, 1 mM EDTA, pH 8.0) was prepared with distilled water and autoclaved at 121°C for 20 minutes. A volume of 100 $\mu$L of amplified DNA taggant (from 4–5 PCR reactions) was introduced into the 1 L carrier buffer and blended on a laboratory blood mixer to achieve complete spatial homogeneity.

## 2.4 Product Integration

- **Commercial Pharmaceutical QR Codes**: The taggant spray formulation was directly applied onto 2D DataMatrix and QR code panels of commercial pharmaceutical packaging (e.g., Macter syrup bottles, Bismol, and tablet cartons).

- **Liquid Matrices**: Directly dosed into liquid consumables (*Gelsemium 30, Surkhana*) at parts-per-billion concentrations without altering color, viscosity, or chemical stability.

- **Industrial Roll-to-Roll Spraying**: Applied via an automated 6-station roll-to-roll spray machine with infrared flash curing (60°C, 1.5s).

## 2.5 Field Recovery

1. **Surface Swabbing**: A sterile swab moistened with recovery buffer wipes the printed QR code or product security zone.
2. **Chemical Etching**: For encapsulated samples, mild fluoride buffer ($NH_4F/HF$) dissolves outer silica nano-coatings within 10 minutes.
3. **Purification**: Standard silica spin column elutes purified DNA in 25.0 $\mu$L nuclease-free water.

## 2.6 qPCR Verification Protocols

- **Probe-Based qPCR Formulation (20 $\mu$L Volume)**: Reagents comprise 4.0 $\mu$L of 2× Super Real Probe Premix, 0.6 $\mu$L forward primer, 0.6 $\mu$L reverse primer, 0.4 $\mu$L fluorophore probe, 1.0–2.0 $\mu$L template DNA, and 12.4–13.4 $\mu$L RNase-free $ddH_2O$.
- **Thermal Cycling & Melt Analysis**: Initial hot-start activation at 95°C for 15 min, followed by 40 cycles of denaturation (95°C, 30s), annealing (55°C, 30s), extension (72°C, 45s), hold at 95°C for 60s, and high-resolution melt acquisition from 72°C to 95°C at 0.3°C/s.
- **4-Channel Deconvolution**: The recovered DNA is run with 4 sequence-specific fluorophore probes (FAM, HEX, Texas Red, Cy5) to simultaneously confirm the 4-probe code against the central cloud ledger.

# 3. National Scale Calculations

## 3.1 Sequence Space & Combinatorial Entropy

For a linear DNA construct of arbitrary length $N$, with 4 canonical nucleotides (A, T, C, G) at each position, the total number of unique sequence combinations scales as $4^N$.

Taking the $N = 200$ base pairs prototype synthesized in our proof-of-concept as an example, the theoretical sequence space $S_{\text{total}}$ is:

$$S_{\text{total}} = 4^{200} = 2^{400} \approx \mathbf{2.58 \times 10^{120}} \text{ combinations} \tag{3.1}$$

This sequence space ($2.58 \times 10^{120}$) is vastly larger than the estimated number of atoms in the observable universe ($\approx 10^{80}$), ensuring that no two randomly generated sequences will ever be identical.

## 3.2 72-Probe Combinations

In actual implementation, the central facility maintains a physical library of $M = 72$ pre-validated, orthogonal 24 bp DNA probes. By selecting $k = 4$ distinct probes per product batch, the total number of unique batch combinations $C(72, 4)$ is:

$$C(72,4) = \binom{72}{4} = \frac{72!}{4!(72-4)!} = \frac{72 \times 71 \times 70 \times 69}{4 \times 3 \times 2 \times 1} = \mathbf{1,028,790} \text{ unique batch codes} \tag{3.2}$$

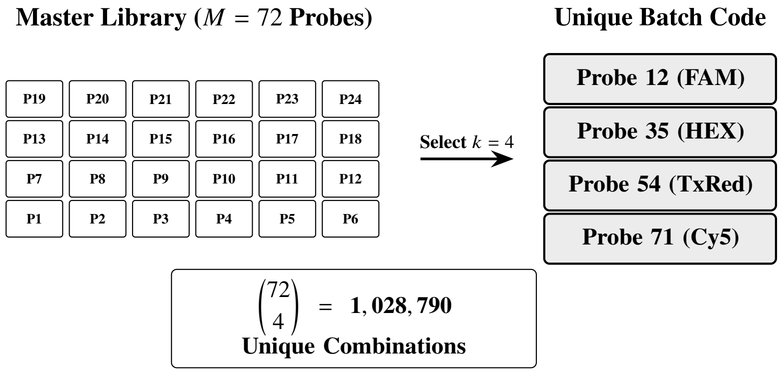


**Figure 3.1.** Combinatorial mixing of 4 probes from a 72-probe library yielding over 1 million unique codes.

This allows the state to generate **over 1 million unique, non-interfering physical batch codes** from a single compact library of 72 vials.

## 3.3 Multi-Sector Application

Table 3.1 illustrates how expanding the library size ($M$) and multiplex depth ($k$) supports nationwide multi-sector scaling.

**Table 3.1.** Combinatorial Scaling and National Industrial Applications

| **Library Size ($M$)** | **Multiplex ($k$)** | **Unique Codes** | **Target National Sector** |
|---|---|---|---|
| $M = 48$ | $k = 4$ | $194,580$ | National Pharmaceutical Verification. |
| $M = 72$ | $k = 4$ | $1,028,790$ | **Multi-Sector**: Pharma, Petroleum, Defense, Tax Stamps. |
| $M = 96$ | $k = 4$ | $3,321,960$ | Consumer Goods & Agrochemicals. |
| $M = 96$ | $k = 5$ | $66,439,200$ | Security Documents & Currency. |
| $M = 120$ | $k = 5$ | $190,578,024$ | Universal Multi-Decade National Scale. |

## 3.4 Dosing & Unit Economics

- **Micro-Dosing**: DNA taggants are used at parts-per-billion (ppb) concentrations (< 1 ng/L). A standard 20 $\mu$L qPCR reaction requires only $\sim$ 10 femtograms $(10^{-14}$ g) of DNA.
- **Batch Yield**: A single 1.0-Liter carrier formulation tags **over 100 million physical units** (bottles, cartons, or tax stamps).
- **Unit Cost**: The chemical reagent cost per tagged unit is **less than PKR 0.01 (< 0.00004 USD)**, making the system economically viable for mass deployment.

# 4. Experimental Results

## 4.1 Overview

Wet-lab experiments conducted at ASAB-NUST successfully achieved:

1. Rapid colony PCR recovery (95°C, 10 min heat-lysis) and gradient PCR optimization (57°C) of the 200 bp prototype construct.
2. Validation of 1.0-Liter bulk spray formulation homogeneity ($C_q \approx 4.2$–$5.7$ across 5 random fractions).
3. Real-time multiplex qPCR detection with 100% optical signal across all 4 probe channels (FAM, HEX, Texas Red, Cy5) in a single reaction.

## 4.2 Gel Electrophoresis & Annealing Optimization

Gradient PCR was evaluated on a 2.0% agarose gel across annealing temperatures from 54°C to 59°C (Figure 4.1).

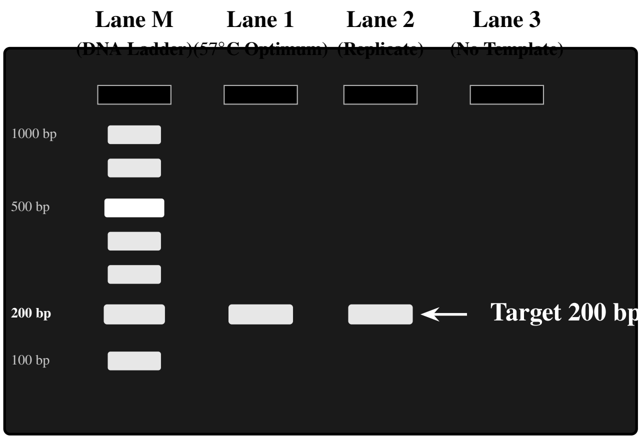


**Figure 4.1.** Agarose gel electrophoresis showing the discrete 200 bp amplicon band with zero non-specific amplification.

The temperature range 55°C–57°C yielded sharp amplification, with 57.0°C identified as the optimum annealing temperature, producing a single, well-defined band at 200 bp without primer-dimers.

## 4.3 Bulk Formulation Homogeneity (1.0 L Reagent)

To evaluate whether DNA taggants remain uniformly distributed within a large-volume liquid carrier, multiple independent fractions (F1 to F7 and product recovery replicates

PR1–PR2) were sampled from the 1.0-Liter TE formulation and evaluated via qPCR (Table 4.1).

**Table 4.1.** qPCR Assessment of Spatial Homogeneity Across 1.0-Liter Bulk Reagent

| **Sample Fraction** | **Template Volume** | **Observed $C_q$** | **Homogeneity Status** |
|---|---|---|---|
| **F1** | 1.0 $\mu$L | **4.21** | Positive (Uniform Dispersion) |
| **F2** | 1.0 $\mu$L | **5.03** | Positive (Uniform Dispersion) |
| **F3** | 1.0 $\mu$L | **5.75** | Positive (Uniform Dispersion) |
| **F4** | 1.0 $\mu$L | **5.34** | Positive (Uniform Dispersion) |
| **F5 (Replicates R1/R2)** | 1.0–2.0 $\mu$L | **5.03 / 6.01** | Positive (Uniform Dispersion) |
| **F6 (Replicates R1/R2)** | 1.0–2.0 $\mu$L | **5.75 / 6.59** | Positive (Uniform Dispersion) |
| **F7 (Replicates R1/R2)** | 1.0–2.0 $\mu$L | **5.34 / 5.36** | Positive (Uniform Dispersion) |
| **PR (Replicates R1/R2)** | 1.0–2.0 $\mu$L | **3.99 / 5.23** | Positive (Product Recovery) |

The tight cycle threshold cluster ($C_q$ = 3.99–6.59) across all 7 fractions and product replicates confirms complete spatial dispersion and stability throughout the 1.0-Liter spray formulation.

## 4.4 4-Channel Multiplex qPCR Verification

Real-time qPCR was executed with simultaneous 4-channel fluorescence acquisition (Figure 4.2).

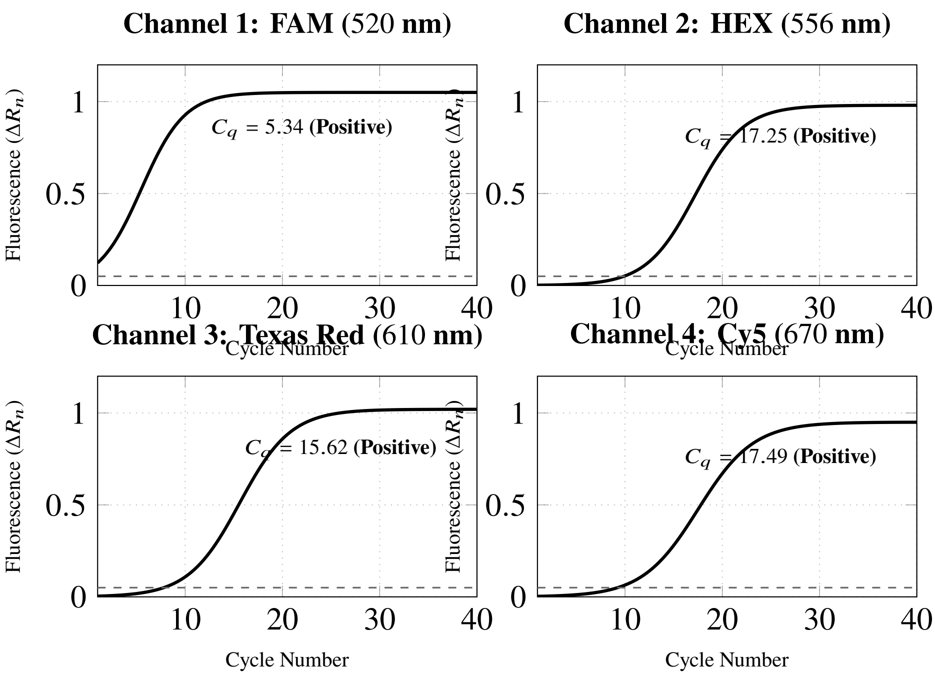


**Figure 4.2.** Simultaneous real-time qPCR curves showing 100% positive detection across all 4 probe channels.

## 4.5 Quantitative Cycle ($C_q$) Data

Table 4.2 presents the experimental quantification cycle ($C_q$) values for single and multiplex probe validation across all tested replicates:

**Table 4.2.** Empirical qPCR Verification Data Across 4 Optical Probe Channels

| Target Probe | Optical Channel | Emission $\lambda$ | Experimental $C_q$ | Verification Status |
|---|---|---|---|---|
| **Probe A** | FAM | 520 nm | **1.29 – 6.01** | Positive (Target Confirmed) |
| **Probe B** | HEX | 556 nm | **8.83 – 17.29** | Positive (Target Confirmed) |
| **Probe C** | Texas Red | 610 nm | **14.64 – 16.89** | Positive (Target Confirmed) |
| **Probe D** | Cy5 | 670 nm | **17.00 – 18.83** | Positive (Target Confirmed) |
| **4-Plex Multiplex** | All 4 Channels | Multi | **11.95 – 15.49** | Positive (Full Deconvolution) |

All four fluorophore probes yielded clean amplification with $C_q < 20.0$, verifying that 4-probe multiplex deconvolution operates with 100% specificity.

# 5. Nanomaterial Encapsulation

## 5.1 Nano-Shielding Rationale

For high-stress applications — such as metallic projectile casting, pyrotechnics, intense UV radiation, or chemical washing — synthetic DNA is protected inside an inorganic core-shell nanoparticle (Figure 5.1).

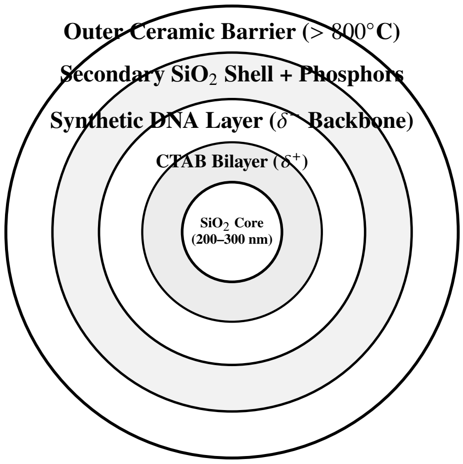


**Figure 5.1.** Multi-layer architecture of the encapsulated DNA taggant particle.

*[Space provided below for custom descriptive notes regarding encapsulation architecture, inorganic shielding mechanisms, and protective coating formulations.]*

## 5.2 Encapsulation Protocols

*[Space provided for step-by-step chemical protocols including Stöber sol-gel synthesis, CTAB functionalization, DNA loading, secondary shell growth, and phosphor doping.]*

### 5.2.1 Stage I: Silica Core Synthesis

### 5.2.2 Stage II: CTAB Functionalization & Charge Inversion

### 5.2.3 Stage III: DNA Loading & Secondary Silica Encapsulation

### 5.2.4 Stage IV: Phosphor Doping & Outer Ceramic Shielding

## 5.3 Characterization Data

*[Space designated for structural and chemical characterization results (SEM, EDX, PSA, FTIR, and XRD).]*

### 5.3.1 SEM & EDX Analysis

### 5.3.2 Particle Size Analysis (PSA)

### 5.3.3 FTIR Spectroscopy

### 5.3.4 XRD Phase Analysis

# 6. Software Ecosystem

## 6.1 System Architecture

The software ecosystem operates on a dual-database architecture connecting four specialized applications (Figure 6.1):

1. **DNA Records Database (Primary Source of Truth)**: A secure Firebase database storing synthesized sequence data, construct length, GC-content, primer docks, and 4-probe fluorophore assignments under the `/dna_exports` node.
2. **Supply Chain Tracking Database**: A live telemetry database structured around `/users`, `/products`, and `/transactions` to record custody handoffs in real time.

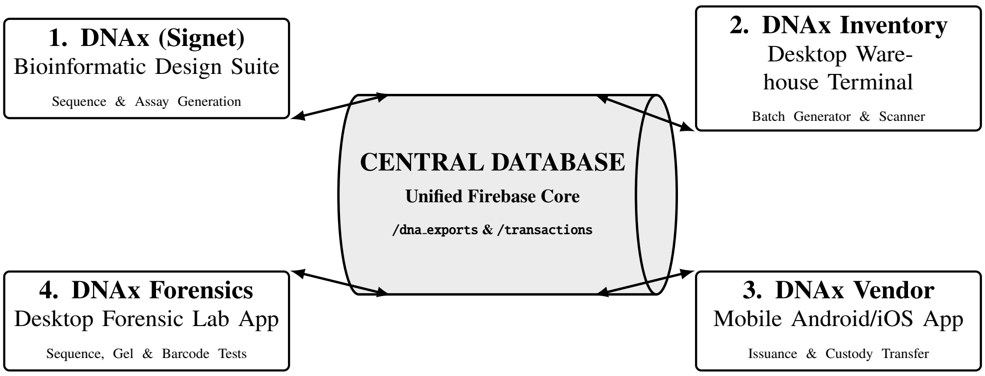


**Figure 6.1.** The 4-app software ecosystem operating on the central database ledger.

## 6.2 Application Modules

### 6.2.1 1. DNAx (Bioinformatic Design Platform)

Deployed as a hybrid web/desktop application (React, Vite, Tailwind CSS, PyWebView, and SQLite/Firebase):

- **Sequence Generator**: Generates orthogonal, custom-length constructs (such as the 200 bp prototype) with constrained GC content (48%–52%) and zero homopolymers.
- **Thermodynamic Solver**: Calculates SantaLucia nearest-neighbor melting temperatures ($T_m$) for primer pairs (58.5°C) and 4 multiplex probes (68.0°C).
- **Safety Screening**: Performs local BLAST screening against offline NCBI genome mirrors.

- **Dossier Export**: Generates molecular specification PDFs embedded with high-resolution 2D QR codes for physical batch tagging.

#### 6.2.2 2. DNAx Inventory (Manufacturing & Warehouse App)

A Python/Tkinter desktop application deployed in production plants and warehouses:

- **Batch Generator (`generator.py`)**: Allocates unique DNA tag identifiers to incoming production runs.
- **Barcode Scanner (`scanner.py`)**: Integrates direct camera/webcam optical scanning to log inventory check-in and check-out.
- **Stock Ledger (`items.py`)**: Tracks live item availability and stock status in the cloud.
- **Audit Trail (`history.py`)**: Records all internal movement logs.

#### 6.2.3 3. DNAx Vendor (Mobile Supply Chain App)

A cross-platform React Native/Expo mobile application for field inspectors, distributors, and retailers:

- **Product Issuance (`issue.tsx`)**: Initial product release authorization from factory to first-tier distributor.
- **Custody Transfer (`transfer.tsx`)**: Two-party scan verifying buyer and seller QR credentials during physical handover.
- **Live Scanner (`scanner.tsx`)**: Real-time camera barcode/QR scanner for instant status queries.
- **Transaction History (`history.tsx`)**: Timestamped chain-of-custody audit trail stored in Firebase (`/transactions`).

#### 6.2.4 4. DNAx Forensics (Laboratory Verification Suite)

A specialized desktop Python/Tkinter testing suite engineered for forensic agencies:

- **Sequence Matching (`sequence_test.py`)**: Ingests Sanger or NGS sequence strings and searches the cloud database (`/dna_exports`) for exact sequence matches.
- **Gel Sizing Test (`gel_test.py`)**: Cross-references electrophoretic base-pair lengths (e.g., 200 bp) against registered company profiles.
- **Barcode Matching (`barcode_test.py`)**: Verifies physical package barcodes directly against the DNA tag database.
- **qPCR Fluorophore Deconvolution**: Matches empirical $C_q$ trajectories (FAM, HEX, Texas Red, Cy5) to issue legal Certificates of Authenticity.

## 6.3 Conclusion

The integration of these four applications establishes a closed-loop digital and physical tracking ecosystem:

- **Unbroken Traceability**: Every physical product tagged with synthetic DNA is mirrored by an active record in the cloud.
- **Multi-Platform Accessibility**: Dedicated interfaces for research scientists (DNAx), warehouse operators (DNAx Inventory), logistics agents (DNAx Vendor), and forensic officers (DNAx Forensics).
- **Full Sovereign Control**: Completely indigenous software architecture free of third-party licensing dependencies.